COURSE TITLE: COMPUTER GRAPHICS

1.1 Introduction to computer graphics, advantages of computer graphics, applications of computer graphics.
1.2 Display devices: Raster scan display, random scan display, Flat panel display, LED display, LCD display, plasma display, touch screen.
1.3 Graphics functions and standards. Types of graphics file formats: General explanation, advantages & disadvantages.
1.4 Latest trends in Computer Graphics. Virtual reality, Augumented reality.
1.5 Co-ordinate systems (2D, 3D): cartesian, Polar.

1.1 Introduction to computer graphics

The computer is an information processing machine. It is a tool for storing, manipulating and correlating data. There are many ways to communicate the processed information to the user. The computer graphics is one of the most effective and commonly used way to communicate the processed information to the user. It displays the information in the form of graphics objects such as pictures, charts, graphs and diagrams instead of simple text. Thus we can say that computer graphics makes it possible to express data in pictorial form. The picture or graphics object may be an engineering drawing, business graphs, architectural structures, a single frame from an animated movie or a machine parts illustrated for a service manual. It is the fundamental cohesive concept in computer graphics. Therefore, it is important to understand -

- How pictures or graphics objects are presented in computer graphics ?
- How pictures or graphics objects are prepared for presentation ?
- How previously prepared pictures or graphics objects are presented ?
- How interaction with the picture or graphics object is accomplished ?

In computer graphics, pictures or graphics objects are presented as a collection of

In computer graphics, pictures or graphics objects are presented as a collection of discrete picture elements called pixels. The pixel is the smallest addressable screen element. It is the smallest piece of the display screen which we can control. The control is achieved by setting the intensity and colour of the pixel which compose the screen. This is illustrated in Fig. 1.1.

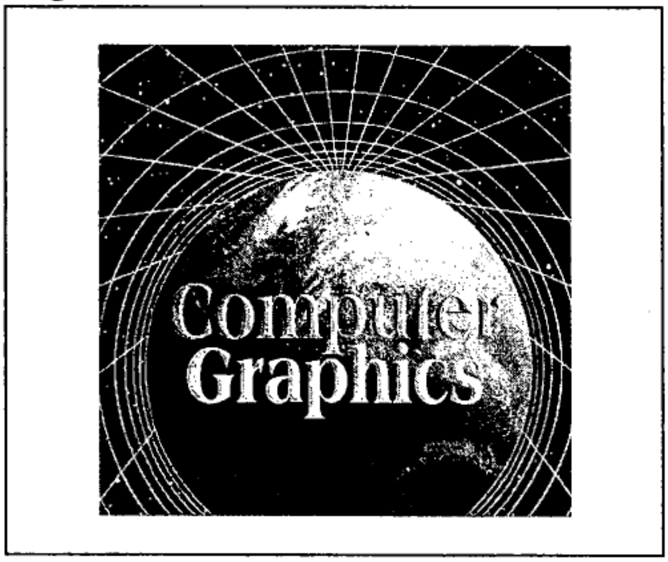

Fig. 1.1 Representation of picture

Each pixel on the graphics display does not represent mathematical point. Rather, it represents a region which theoretically can contain an infinite number of points. For example, if we want to display point P_1 whose coordinates are (4.2, 3.8) and point P_2 whose coordinates are (4.8, 3.1) then P_1 and P_2 are represented by only one pixel (4, 3), as shown in the Fig. 1.2. In general, a point is represented by the integer part of x and integer part of y, i.e., pixel (int (x), int (y)).

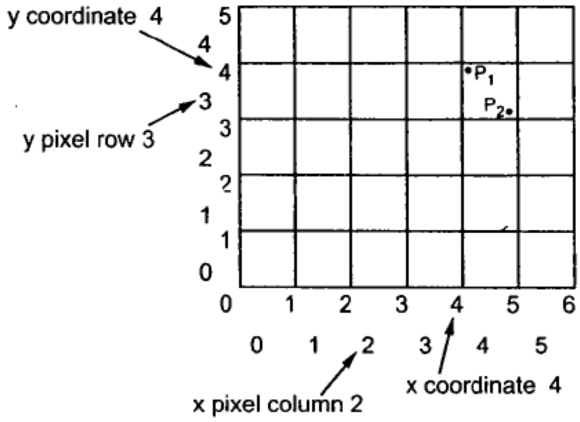

Fig. 1.2 Pixel display area of 6 × 5

Application of Computer Graphics

Computer Graphics has numerous applications, some of which are listed below:

Computer graphics user interfaces (GUIs) – A graphic, mouse-oriented paradigm which allows the user to interact with a computer.

Business presentation graphics - "A picture is worth a thousand words".

 Cartography - Drawing maps.

 Weather Maps – Real-time mapping, symbolic representations.

 Satellite Imaging - Geodesic images.

 Photo Enhancement - Sharpening blurred photos.

 Medical imaging - MRIs, CAT scans, etc. - Non-invasive internal examination.

 Engineering drawings - mechanical, electrical, civil, etc. - Replacing the blueprints of the past.

 Typography - The use of character images in publishing - replacing the hard type of

the past.

 Architecture - Construction plans, exterior sketches - replacing the blueprints and

hand drawings of the past.

 Art - Computers provide a new medium for artists.

 Training - Flight simulators, computer aided instruction, etc.

 Entertainment - Movies and games.

 Simulation and modeling - Replacing physical modeling and enactments

Some of the applications of computer graphics are:

Computer Art:

Using computer graphics we can create fine and commercial art which include animation packages, paint packages. These packages provide facilities for designing object shapes and specifying object motion.Cartoon drawing, paintings, logo design can also be done.

Computer Aided Drawing:

Designing of buildings, automobile, aircraft is done with the help of computer aided drawing, this helps in providing minute details to the drawing and producing more accurate and sharp drawings with better specifications.

Presentation Graphics:

For the preparation of reports or summarising the financial, statistical, mathematical, scientific, economic data for research reports, managerial reports, moreover creation of bar graphs, pie charts, time chart, can be done using the tools present in computer graphics.

Entertainment:

Computer graphics finds a major part of its utility in the movie industry and game industry. Used for creating motion pictures , music video, television shows, cartoon animation films. In the game industry where focus and interactivity are the key players, computer graphics helps in providing such features in the efficient way.

Education:

Computer generated models are extremely useful for teaching huge number of concepts and fundamentals in an easy to understand and learn manner. Using computer graphics many educational models can be created through which more interest can be generated among the students regarding the subject.

Training:

Specialised system for training like simulators can be used for training the candidates in a way that can be grasped in a short span of time with better understanding. Creation of training modules using computer graphics is simple and very useful.

Visualisation:

Today the need of visualise things have increased drastically, the need of visualisation can be seen in many advance technologies , data visualisation helps in finding insights of the data , to check and study the behaviour of processes around us we need appropriate visualisation which can be achieved through proper usage of computer graphics

Image Processing:

Various kinds of photographs or images require editing in order to be used in different places. Processing of existing images into refined ones for better interpretation is one of the many applications of computer graphics.

Machine Drawing:

Computer graphics is very frequently used for designing, modifying and creation of various parts of machine and the whole machine itself, the main reason behind using computer graphics for this purpose is the precision and clarity we get from such drawing is ultimate and extremely desired for the safe manufacturing of machine using these drawings.

Graphical User Interface:

The use of pictures, images, icons, pop-up menus, graphical objects helps in creating a user friendly environment where working is easy and pleasant, using computer graphics we can create such an atmosphere where everything can be automated and anyone can get the desired action performed in an easy fashion.